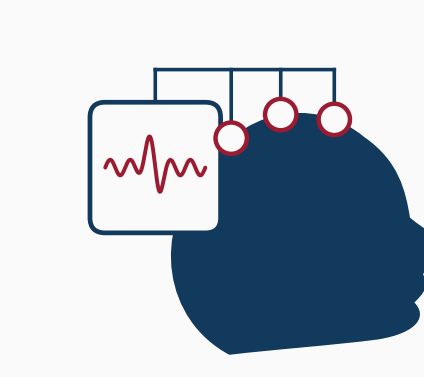


NeuroState: Gradient-Isolated Boundary Learning for Interpretable EEG Transformers

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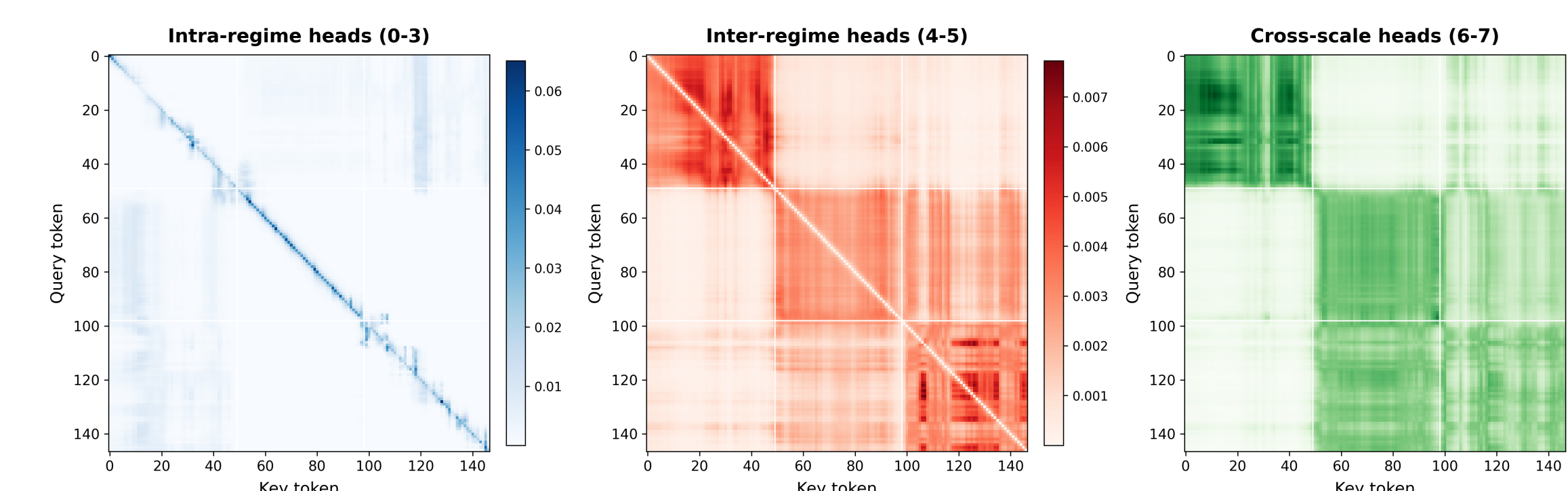
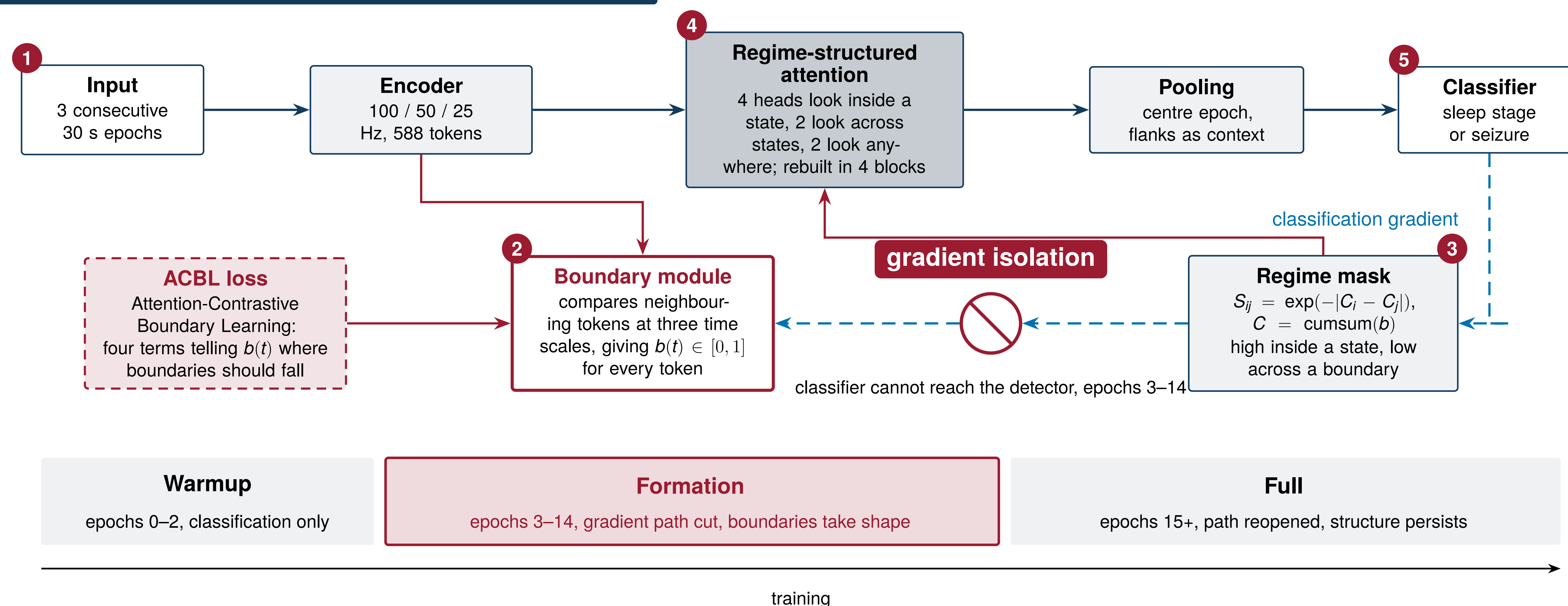
Code

1 EEG transformers report *which* state the brain is in, not *when* it changes. Train a boundary head to supply the timing and it **collapses to a near-constant output**.

2 **Gradient isolation**, cutting the classification gradient into the boundary head for epochs 3–14, is the only fix among eleven tried.

3 A falling loss and a rising accuracy are **not evidence** that a jointly trained second head learned anything.

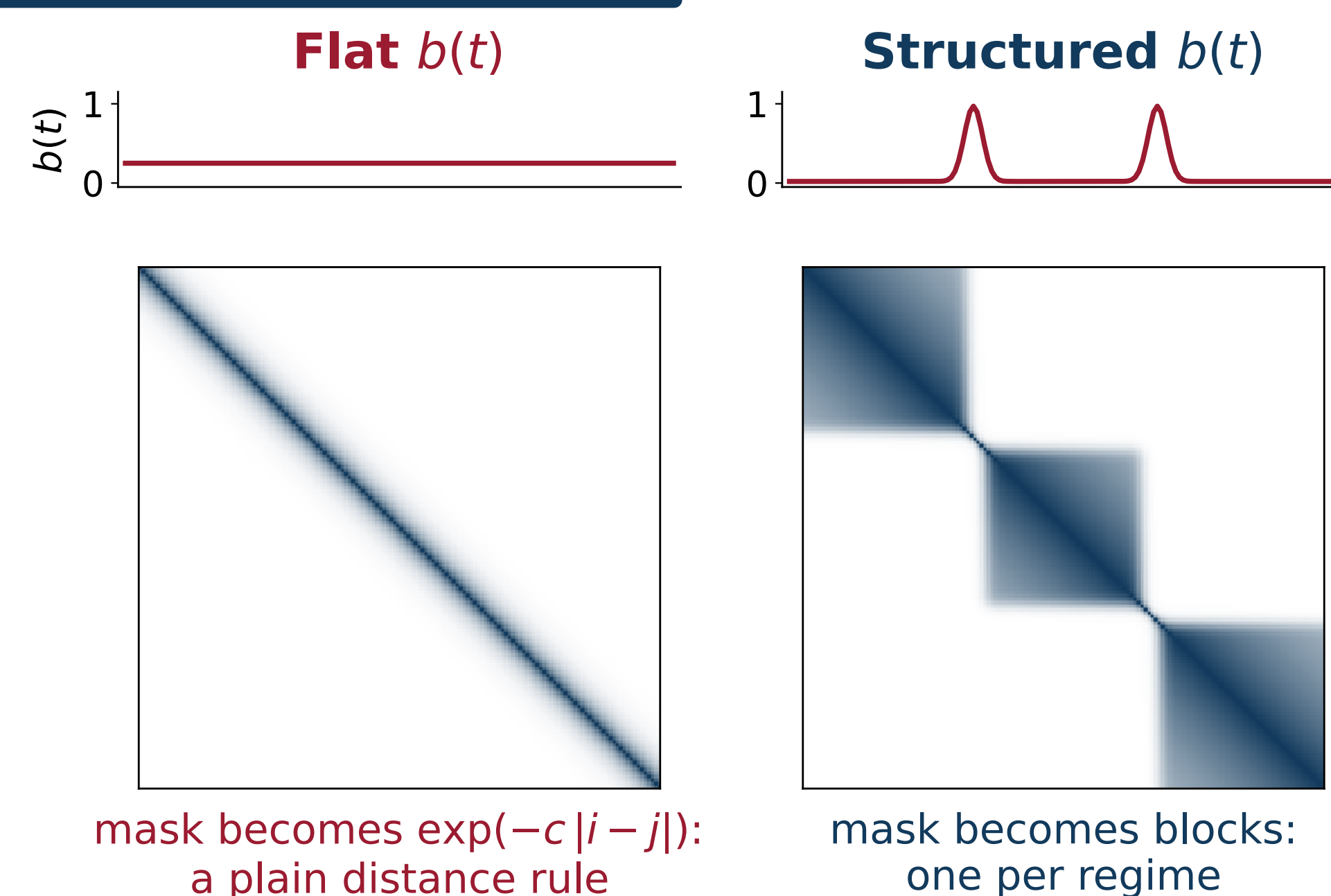
1 · Method, and the attention it produces



• One Sleep-EDF window, N3 → N2 → N3. Intra-regime heads hold the block diagonal, inter-regime heads the off-diagonal blocks, cross-scale heads stay open.

$$b(t) = \sigma(W_f[d_1, d_4, d_{16}]^T + b_f) \quad (\text{boundary probability at every token})$$
$$C_i = \sum_{j \leq i} b(j), \quad S_{ij} = \exp(-|C_i - C_j|) \quad (\text{regime mask: high within a state})$$
$$\tilde{a}_{ij}^{(h)} = a_{ij}^{(h)} + \log(M_{ij}^{(h)} + \epsilon) \quad (\text{mask enters attention as a log bias})$$
$$M = S \begin{pmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 4 & 4 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} \quad (\text{intra-regime, inter-regime, cross-scale})$$

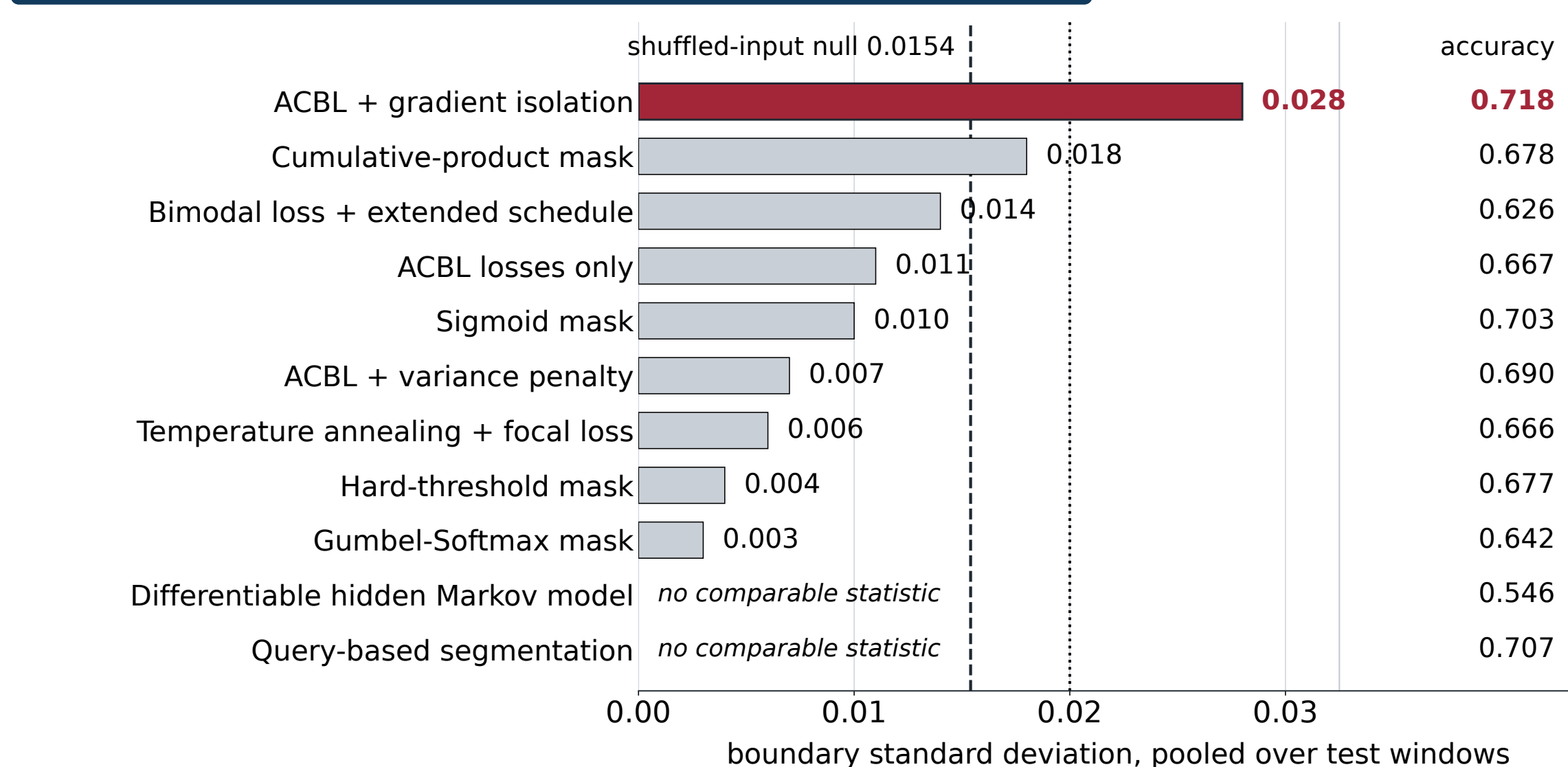
2 · Why it collapses



$$b \equiv c \Rightarrow S_{ij} = \exp(-c|i-j|) \text{ (a flat detector leaves pure distance decay)}$$

- The mask becomes a distance rule, blind to the signal.
- That rule alone is worth 0.703 accuracy, so the classification gradient into the detector vanishes.
- Nothing flags it. The loss keeps falling and accuracy keeps rising while the detector sits at the degenerate solution.
- Ablation: removing the Gaussian targets flattens it completely, to 0.0001 on Sleep-EDF and 0.001 on CHB-MIT.
- **So collapse has to be tested for.** It takes a statistic on $b(t)$ and a null to compare it against: the shuffled-input null here is $\sigma_b = 0.0154$.

3 · Eleven fixes, one survivor



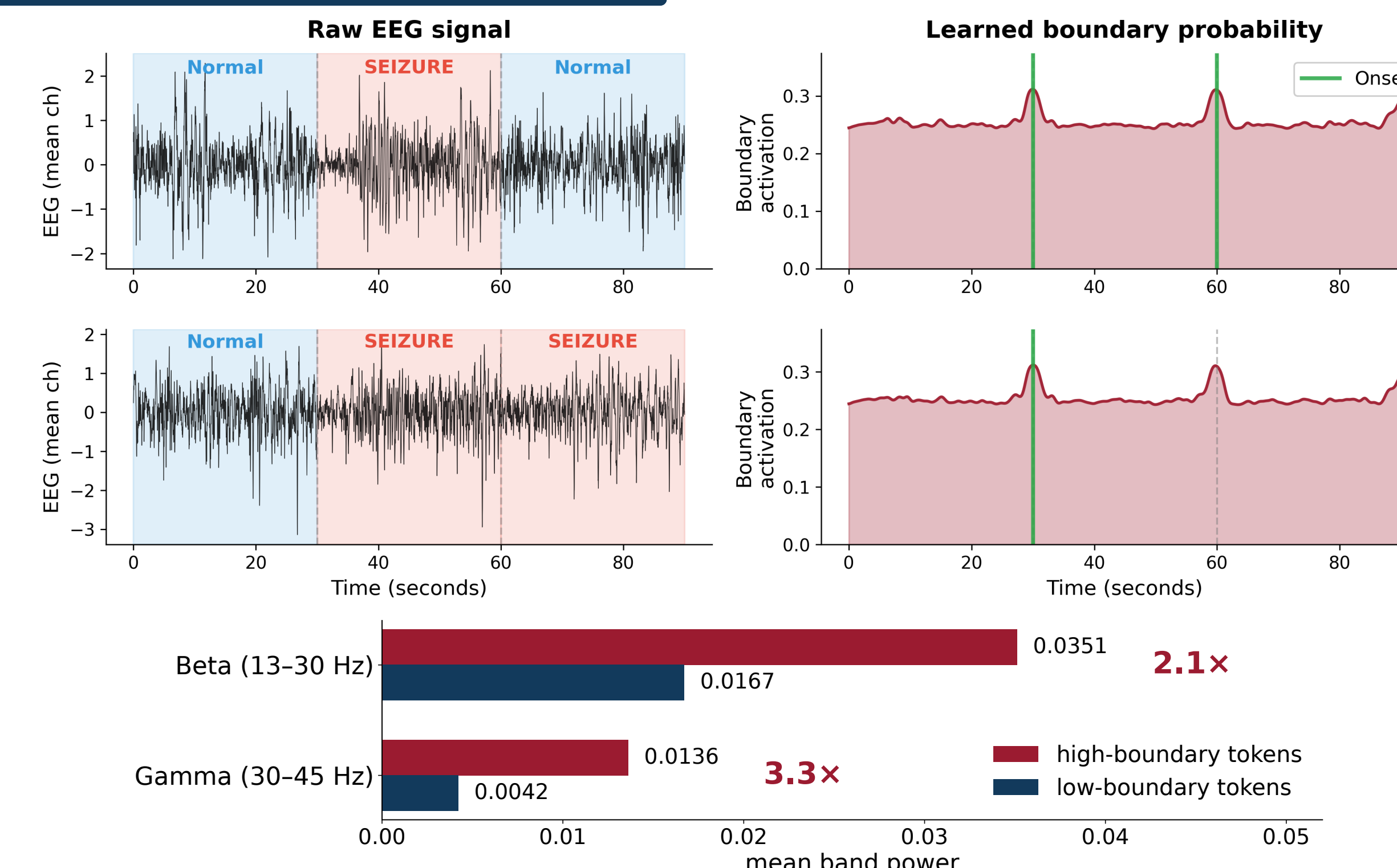
$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{ACBL}}^{\text{detach}} \quad (\text{epochs 3–14: classifier cannot reach } b(t))$$
$$\mathcal{L}_{\text{ACBL}} = \alpha \mathcal{L}_{\text{pseudo}} + \beta (\mathcal{L}_{\text{Gauss}} + \gamma \mathcal{L}_{\text{KL}}) + \delta \mathcal{L}_{\text{var}} \quad (\alpha, \beta, \gamma, \delta = 0.3, 0.5, 0.5, 1.0; \lambda = 0.3)$$

- **0.028** boundary std at **0.718** accuracy. The other eight with a comparable statistic all fall between 0.003 and 0.018, below the 0.020 line that labels a degenerate distribution.
- The same collapse appears under sigmoid masking, discretised masking, a differentiable HMM and query-based segmentation, so it belongs to the joint objective, not to how boundaries are written down.

Dataset	Acc	AUROC	κ	σ_b
Sleep-EDF	0.698	—	0.596	0.026
CHB-MIT	0.901	0.924	0.356	0.014
TUAB [†]	0.916	0.913	0.545	0.003

Five seeds (42–46), a fixed split. [†]single seed.

4 · Is the structure real?



- **Sleep-EDF: yes.** Clears the null in 5/5 seeds, 1.47–1.94×
- **CHB-MIT: no.** Transition windows ($n = 59$) test no different from the rest ($n = 1,437$), $p \geq 0.21$, $d = -0.25 \pm 0.20$. The centre epoch scores above both neighbours in 98.9% of seizure windows and 99.9% of the 1,443 without, so the peaks track epoch joins. Separately, the nearest peak sits a median 30.0 s from onset against 10.5 s for PELT.
- **TUAB: flat by design.** $\sigma_b = 0.003$: whole-recording labels, so there is no within-window change to find.
- **Yet they do carry spectral signal (above).** On Sleep-EDF, high-boundary regions show 2.1× the beta and 3.3× the gamma power of low-boundary regions, $p < 10^{-22}$, consistent with spectral change at a stage transition.
- The interpretability result is preliminary.

5 · What this means

- **Boundary learning:** a secondary head can appear trained while its output is degenerate. Verify it against a null, not against the loss curve.
- **Optimisation:** when the primary task does not need the secondary one, the joint objective has a cheap degenerate solution. Cutting one gradient path for twelve epochs escapes it.
- **Transformers:** a learned signal can shape attention, but only if it forms before the classifier can exploit it.

- **Clinical EEG:** the transition time matters as much as the class label, the sleep stage or seizure call, and a boundary claim needs a control before it is called physiological.
- **Sequence models:** the failure comes from training two jobs together rather than from any particular way of writing boundaries down. Whether it generalises beyond the EEG settings measured here is untested.